

ASSESSMENT FORM

Course: COMP6049001 - Algorithm Design and Analysis

Method of Assessment: Case Study

Semester/Academic Year : 3/2025-2026

Name of Lecturer :

Date :

Class :

Topic : Review II

Group Members :	1
	2
	3
	4
	5
	6
	7
	8

Student Outcomes:

(SO 1) Mampu menganalisis masalah komputasi yang kompleks dan mengaplikasikan prinsip komputasi dan keilmuan lain yang sesuai untuk mengidentifikasi solusi

Able to analyze a complex computing problem and to apply principles of computing and other relevant disciplines to identify solutions

(SO 2) Mampu merancang, mengimplementasikan, dan mengevaluasi solusi berbasis komputasi untuk memenuhi serangkaian persyaratan komputasi dalam konteks ilmu computer

Able to design, implement, and evaluate a computing-based solution to meet a given set of computing requirements in the context of computer science.

Learning Objectives:

(LObj 1.1) Mampu menganalisis masalah komputasi yang kompleks

Able to analyze a complex computing problem

(LObj 2.3) Mampu mengevaluasi solusi berbasis komputasi untuk memenuhi serangkaian persyaratan komputasi tertentu dalam konteks ilmu komputer

Able to evaluate a computing-based solution to meet a given set of computing requirements in the context of computer science

No	Assessment criteria	Weight	Excellent (85 - 100)	Good (75-84)	Average (65-74)	Poor (0 - 64)	Score	(Score x Weight)
1	Ability to solve problems in INC 2025 during the event.	50%	Able to solve > 2 problems in INC 2025 during the event.	Able to solve 2 problems in INC 2025 during the event.	Able to solve 1 problem in INC 2025 during the event.	Present at INC 2025, tried but not able to solve any problems during the event. Not present or plagiarism* results in 0.		
2	Ability to write an analysis of the solutions used in INC 2025.	30%	Able to explain and analyze the complexity of the algorithms for solved problems and provide a rigorous proof of the correctness of the algorithm (max. 3).	Able to explain and analyze the complexity of the algorithms for solved problems (max. 3).	Able to provide the source code for the solved problems (max. 3).	Able to provide the source code for the solved problems. Not present results in 0.		
3	Ability to upsolve the unsolved problems in INC 2025 after the event.	20%	Able to explain the correct solution for at least 1 unsolved problem (if exist) and provide the source	Able to explain the correct solution for at least 1 unsolved problem (if exist). Report on why your	Explain on why your group cannot solve the problem during event. Explain your	Explain on why your group cannot solve the problem during event.		

No	Assessment criteria	Weight	Excellent (85 - 100)	Good (75-84)	Average (65-74)	Poor (0 - 64)	Score	(Score x Weight)
			code. Report on why your group cannot solve the problem during event.	group cannot solve the problem during event.	attempt during the event and why it failed.			
	Total Score: $\sum(\text{Score} \times \text{Weight})$							

Remarks:

***Plagiarism** here means that student's code has high similarity to others.

ASSESSMENT METHOD

Instructions

Before INC 2025

1. Students form a group consisting of 3 people.
2. Follow the provided instruction to register to INC 2025.

During INC 2025

1. Follow the schedule and rules of INC 2025.
2. Solve as many problems as your team could solve. The number of solved problems will be used in assessment criteria 1.
3. If the students are not present during the event, then the score for criteria 1 & 2 automatically goes to 0 (zero).
4. If the students are caught cheating for certain problems (have high similarity to other student's code), then their answer will be nullified (which affects criteria 1). This rule applies to the cheater and the giver. The decision is final and cannot be disturbed/bargained.
5. Write your own code, as the internet or AI may give similar code with other students and nullify your answer.

After INC 2025

1. Write a problem analysis (editorial) for the problems you solved. If you solve > 3 problems during INC, pick maximum 3 of the solved problems. The way you analyze and explain your solution will be used in assessment criteria 2.
2. Take your time to keep practicing by solving the unsolved problems during INC 2025. You are allowed to discuss the problems with other groups, but your group must show that you understand the solution and be able to solve the problems. If you upsolve > 1 problems, pick at least 1 of the upsolved problems. The way you explain the solution will be used in assessment criteria 3.

Note: if you solved all problems during the event, you don't need to do anything for assessment criteria 3.

Project Output

- Your participation in INC 2025
- Problem analysis (editorial) for the solved problems during INC 2025
- Explanation (and source code) of the upsolved problems after INC 2025

Note for Lecturers:

1. Students submit their project output in the last week of lecture
2. Please verify the project output with the INC 2025 scoreboard